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SIXTH SEMESTER B.Tech/B.E UNIVERSITY EXAMINATION, JUNE – 2016

INTRODUCTION TO FINITE ELEMENT METHODS

Time: 3 Hours **Maximum Marks:70**

Instructions:

1. Answer any **FIVE** full Questions.
2. Missing data may be suitably assumed.

1.	**a.**	What in meant by finite element method, explain in the context of continuum mechanics.	**3.5**
	b.	What are shape functions in FEM? Discuss. Show it is used in nodal variables.	**3.5**
	c.	Discuss closed form, approximation solution and FEM solution in structural analysis, with an example.	**3.5**
	d.	Explain 1D, 2D and 3D bar element, truss element , plate and solid elements with example in FEM.	**3.5**
2.	**a.**	Explain total potential energy in FEM. Discuss various types of forces in a continuum under loading. What is minimum potential energy?	**3.5**
	b.	What is plane-stress and plane-strain problem? Write stress-strain relation matrix for the two cases.	**3.5**
	c.	Explain local, global and natural coordinate system in FEM, with example.	**3.5**
	d.	Explain equilibrium, compatibility equations and types of boundary conditions in FEM.	**3.5**
3.	**a.**	For a linear spring as shown in Fig. 3a, derive the load-displacement relation using energy method.	**7.**

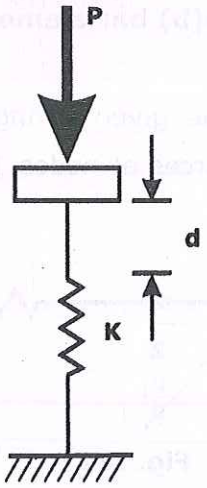


Fig. 3(a)

b. Determine the displacements of at the nodes of the spring system shown in Fig 3b. **07**

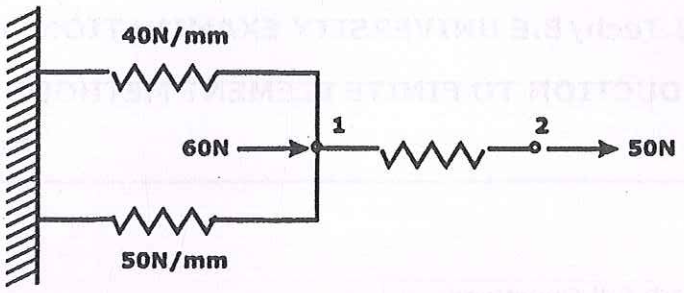


Fig. 3(b) Spring system

4. a. For a two noded bar element shown in fig.4a. Caritesian coordinate system, determine ξ (in natural coordinate system) and shape functions $N_1(\xi)$ and $N_2(\xi)$ at node 1 and 2 respectively. **07**

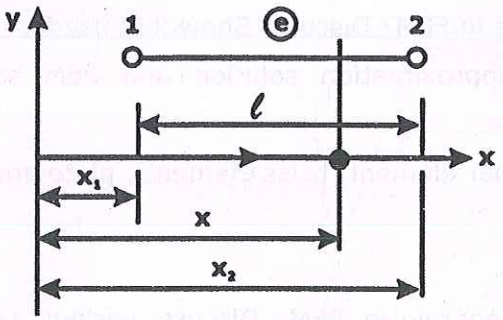


Fig. 4(a)

b. The bar element shown in fig.4b has an axial displacement field u, linear in x and is dependent on nodes D.O.F u_1 and u_2. Determine ξ, N_1 and N_2, with given displacements u_1 =0.5 mm and u_2=0.7 mm. Evaluate nodal displacement at point P. **07**

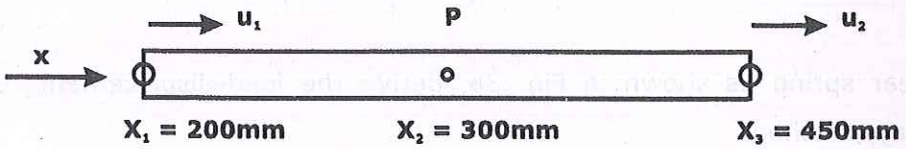


Fig. 4(b) bar element

5. a. Derive the stiffness matrix for the given spring system whose u's are nodal displacements and F's are nodal forces at nodes 1,2 and 3. K_1 and K_2 are spring stiffnesses. **07**

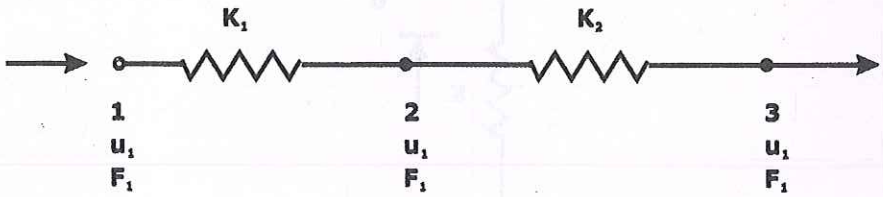
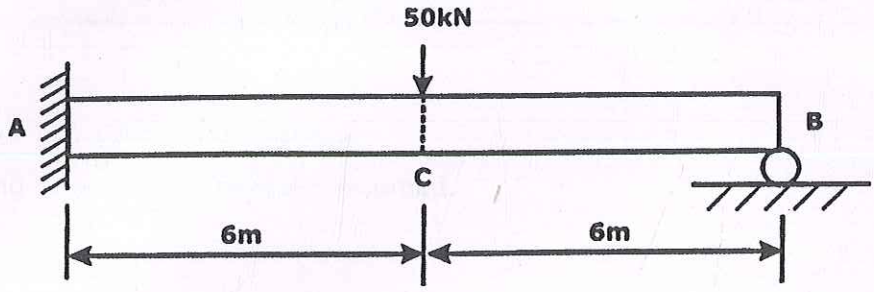


Fig. 5(a)

b. For the given spring system in Fig.5a, consider u_1=0 and F_2 =F_3 =P, find displacements u_2 and u_3 and reaction force at F_1 node 1. **07**

6. a. Explain flexibility matrix method in FEM, Maxwell's reciprocal theorem, compatibility equation. **04**

b. Determine reactions at all supports and moment at the fixed end of the given beam structure, using flexibility method. **10**



Assume the beam is of solid uniform cross section and EI. Draw the shear force and bending moment diagram.

Hint: Deflection of cantilever beam $= \frac{PL^3}{3EI}$, $slope\ at\ c = \frac{PL^2}{2EI}$.

7. a. Explain Gauss elimination technique to solve liner simultaneous equation of form [A] {X}={b}. **04**

b. Solve the given linear simultaneous equations: **10**

$$x_1 - 2x_2 + 6x_3 = 0$$
$$2x_1 + 2x_2 + 3x_3 = 3$$
$$-x_1 + 3x_2 \qquad = 2$$

Using Gauss elimination technique

8. a. What are the general constituents of a FEM software. **3.5**

b. Name any 3 FEM softwares popular in the industry. **03**

c. Explain preprocessing and post processing in FEM analysis. **3.5**

d. Explain eigen value problem. Discuss briefly its application to structural design with an example from aerospace structure. **04**